A few thoughts on gold standard science

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Gold Standard Science?

White House Executive Order



THE WHITE HOUSE
WASHINGTON

PRESIDENTIAL ACTIONS

Restoring Gold Standard Science

Executive Orders May 23, 2025

By the authority vested in me as President by the Constitution and the laws of the United States of America, including section 7301 of title 5, United States Code, it is hereby ordered: Section 1. Policy and Purpose. Over the last 5 years, confidence that scientists act in the best interests of the public has fallen significantly. A majority of researchers in science, technology, engineering, and mathematics believe science is facing a reproducibility crisis. The falsification of data by leading researchers has led to high-profile retractions of federally funded research.

Source: https://www.whitehouse.gov/presidential-actions/2025/05/restoring-gold-standard-science/

What are people concerned about?

Metascience literature has been warning about low standards in research for decades.

My reading of most-cited papers and online discourse:

- Publication bias is rife
- p-hacking is rife (anything can be made significant!)
- “Most published results are false”
- Significant results do not replicate
- Some domains are bad, some are more gold standard
- We should use p-value of 0.01 or 0.001
- Most studies are “cursed” by low power

…plus concerns about data fraud, paper mills, difficulty in publishing nulls, and more.

Replications are seen as a diagnostic for most problems above.

Replication are top priority in the NIH GSS list



Source: https://www.nih.gov/sites/default/files/2025-08/2025-gss.pdf

What we learned from systematic replications

- Open Science Collab. 2015: 97 psychology studies reporting significant results
  - **36%** replications statistically significant in the original direction
  - mean effect size fell 50%
- Many Labs 2 (Klein et al 2019)
  - **15/28** statistically significant in the same direction
  - focused on comparing replications across many different groups
  - aimed for high power
  - median effect size fell by 75%
- Camerer et al 2018: 21 social science papers from Nature and Science
  - **13/21** statistically significant in the same direction
  - effect sizes fell by 50%
  - aimed for high power (5x sample sizes)
  - also demonstrated that replications can be predicted
- Reproducibility Project: Cancer Biology
  - low feasibility: **only 1 in 4 experiments could be ran!**
  - 0/193 had sufficient information to re-run the experiment without help from authors
  - **about half that ran were successes**

Questions I want to ask in this presentation

- Is there a science-wide “replication crisis”?
- How bad are things currently?
- In which ares?
- What is not working? (which concerns are justfied)
- Of parts that don’t work, which can be improved?

I’ll try to give a few hints—far from complete picture.

”Trust In Science”?

Evidence from 100,000 studies

Benchmarks of Empirical Accuracy in Research

Repository (open source!): https://github.com/wwiecek/BEAR

- BEAR is a "meta"-database of z-values, effect sizes, and standard errors assembled from existing public databases across scientific disciplines.
- Does not contribute new primary data. All credit due to authors of metascience publications and databases from which BEAR pulls.
- Includes millions of z-values, including about 100,000 studies from curated datasets.



Source: https://github.com/wwiecek/BEAR

We analyse BEAR data in a new statistical modeling paper

"Statistical case for (qualified) scientific optimism", with Erik van Zwet (LUMC) and Andrew Gelman (Columbia Uni)

- Input: a set distribution of empirical estimates from papers
- Output: estimates of statistical power, replication probabilities, and more
- Repeat the above for many sets of data
- This part of the presentation is mainly my own interpretation of our (work in progress) paper!

Two main statistical concepts: z-value and SNR

For study i, true effect β_i:

$$z_i = \frac{\hat{\beta}_i}{SE(\hat{\beta}_i)}, \qquad \text{SNR}_i = \frac{\beta_i}{SE(\hat{\beta}_i)}.$$

Assuming an unbiased estimator with approximately normal sampling error,

$$z_i = \text{SNR}_i + \varepsilon_i, \qquad \varepsilon_i \sim N(0,1).$$

Relationship of z to p-values:

$$p = 0.05 \Rightarrow |z| = 1.96, p = 0.01 \Rightarrow |z| = 2.58, p = 0.001 \Rightarrow |z| = 3.29$$

Empirical estimates “production function”

(We will illustrate this in a moment.)

1. Researchers pick a hypothesis (“draw effect size”)
2. Study is designed (fix SE; “aim for 80% power”)
3. Therefore SNR is now fixed too
4. Experiment is conducted, adds sampling variation (z = SNR + N(0,1))
5. Researchers may p-hack (move some z-values around)
6. Some non-significant results (p > 0.05, i.e. |z| < 1.96) may go in the file drawer

If we can model distribution of z-values, we can model underlying SNR! If we know SNR, we can derive properties of studies.

What we estimate

For a study picked at random from a given set:

- **Assurance** (power): probability that $|z| > 1.96$
- **Replication** probability: significant result in the same direction (under same study design)
- **Correct sign** probability: sign(z) = sign(β)

"picked at random" is a useful concept: should you trust an average study? can also condition on $|z|$, e.g. "if a study reported $p = 0.05$, probability of replication is ..."

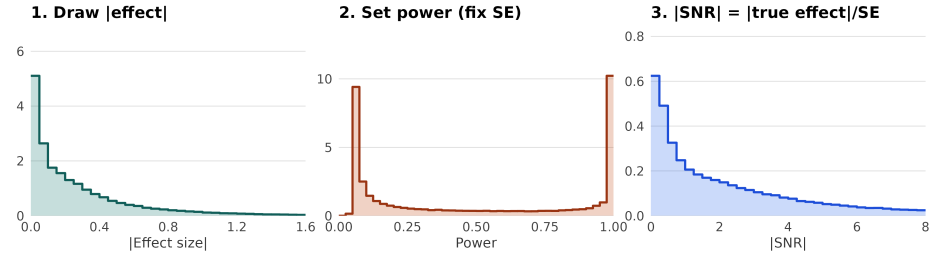
Data Used In The Paper

Paper sample: 11 curated datasets and 80,000 studies across clinical trials, health, ecology and evolution, economics, psychology, psychotherapy, education, and political science.

Dataset	Domain	N studies	k	% signif.
Cochrane	Health and medicine	30,306	1.3	31%
ctgov + EU CTR	Clinical trials	24,468	2.0	48%
Costello and Fox	Ecology and evolution	12,927	6.8	42%
Arel-Bundock et al	Political science	2,252	7.4	47%
Bartos et al	Exercise and cognition	2,239	1.0	26%
Askarov et al	Economics	1,913	11.2	52%
Nuijten et al	Intelligence research	1,913	1.3	53%
Metapsy	Psychotherapy	1,494	2.9	48%
WWC	Education	1,408	8.6	33%
psymetadata	Psychology	721	11.8	36%
Brodeur et al	Economics	176	47.9	37%

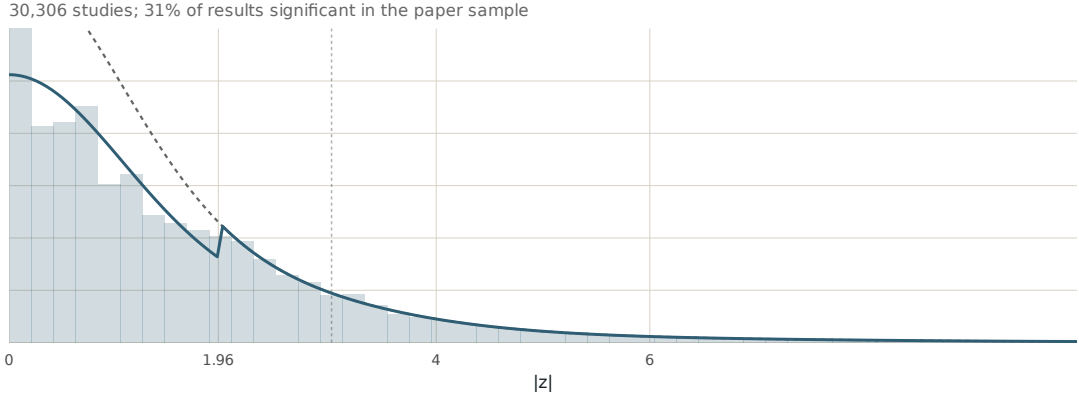
Note: BEAR currently includes 20 datasets and 11.5 million z-values. This slide shows only the 11 curated datasets used in the paper, with counts and percentages taken from the paper table.

Illustrating production function



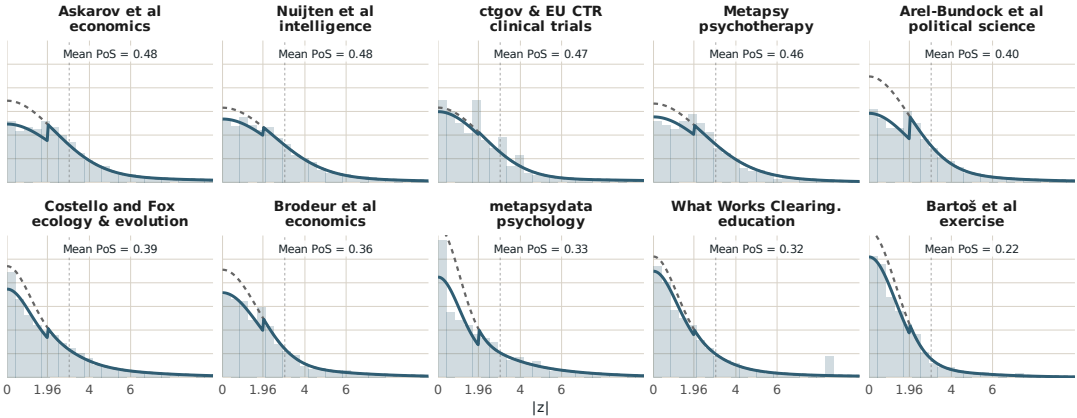
illustrative distributions only, but with shapes based on BEAR datasets

Cochrane Database of Systematic Reviews

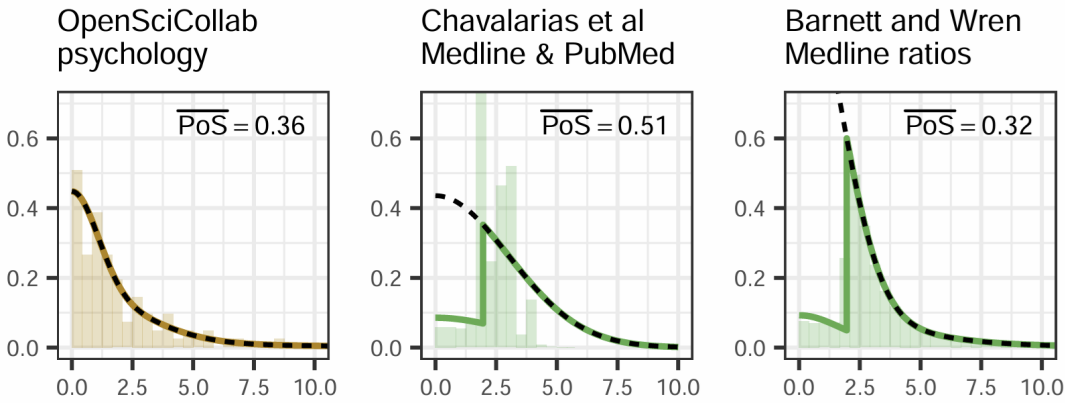


Largest paper dataset, 30,000 studies. Mean assurance = 0.23. Note a decreasing density of absolute z-values with a visible discontinuity at $|z| = 1.96$.

The other 10 curated datasets



Constrast with replications and scraped PubMed values



No discontinuity in OSC 2015. Huge gaps in PubMed!

Main results

Across 11 datasets (and various subsets we tested):

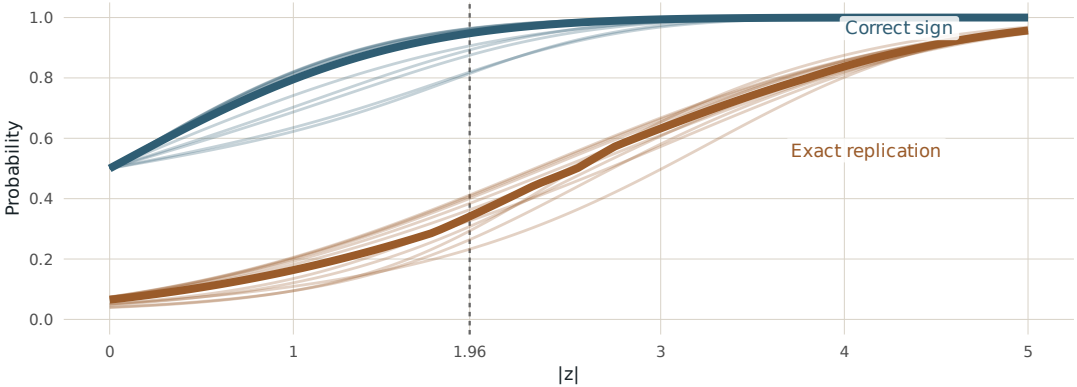
- No large discontinuity at |z| = 1.96 in curated datasets
- Density decreases in |z|
  - This is crucial: the most common scenario is low power/small effects.
  - Even in best domains! (e.g. large Phase 3 blinded clinical trials)
  - What differs across domains is width of the distribution
- Expected probability of...
  - ...replication: 20-50%
  - ...getting the sign right: 70-90%
  - this is for any study (not significant)
- Small proportion of studies reach 80% statistical power (8-29%)

Small studies are not the main issue

- If you have 5% significance and 80% power, 2x sample size would give you 0.2% significance
  - “aiming for p < 0.002” reform
- But you don’t know power in your study!
- Inferred distribution of power in datasets is bimodal. “Good studies” don’t need more subjects, “bad studies” don’t benefit.
- Impact of doubling all samples:
  - **replication probability goes from 21-47% to 25-57%**
  - this would still great at decreasing rate of “false positives”

...the issue is that researchers don’t have a crystal ball!

Sign And Replication Probability



Thin lines are the 11 curated datasets; thick lines are the median curves. Even at $p = 0.001$ ($|z| = 3.29$) replication probability is 0.75 at most.

What have we learned?

(my interpretation!)

1. Dichotomisation of results into true and false is a bad model of reality
2. Even worse if dichotomisation is “significant = true”
3. Statistical significance is not a good proxy for strong result
4. Replication rates will likely remain low (significance twice in a row is hard)
5. … but studies carry useful infrmation!
6. Even a random study highly likely to point in the right direction
7. File drawer problem is less in well-defined research questions
8. Data scraped from PubMed is not fit for assessing empirical research
9. Even best domains (e.g. large scale clinical trials) have same crucial features

Possible implications for metascience

Useful

Significant results do not replicate

Most studies are “cursed” by low power

Not useful

“Most published results are false”

Some domains are bad, some are more gold standard

We should use p-value of 0.01 or 0.001

Don’t know

Publication bias is rife par p-hacking is rife

Replications can be great

You keep using that word...

A few remarks on replication programs

Flavours of replication

- ”Replication” is not well defined. Three simple distinctions.
- **Analytic:** Repeat the analysis using original code and data, e.g. economics.
- **Recipe:** Do the experiment, but follow the original protocol as closely as possible, e.g. cancer biology.
- **Conceptual:** Test the same claim with updated materials or a new context, e.g. psychology replications.
- Clemens (2015) counted 41 definitions in social science, most of them about robustness.



Flavours of replication, cont'd

- Different questions call for different checks
  - Analytic replication checks for basic errors
  - Recipe replication clarifies tacit knowledge/assumptions
  - Conceptual replication help understand external validity
- The boundaries are not clean.
  - Re-running the code but switching to a new implementation of the same method
  - Re-running the same program in the same place, but five years later
- Differences pretty obvious to practicioners & usually easy to choose…
- …but "failure to replicate" conveys little information

What counts as failure?

- Failure is sometimes close to binary.
  - If the code does not run, the data cannot be produced, or an image is manipulated, confidence should fall sharply.
- Most empirical disputes are less clean.
  - An original estimate of `d = 0.30, p = 0.001` and a replication estimate of `d = 0.15, p = 0.10` can imply partial corroboration, a smaller effect, or “failure”, depending on the rule.
- Significance alone is too crude a standard.
  - Success could mean the same sign, similar magnitude, overlapping uncertainty, or persuasive pooled evidence.
- The useful agenda is broader than “more replications”.
  - The goal is to increase confidence in findings that are real and useful, and to discard fragile claims more quickly.

“Non-replication” without replication

- Not considered replication:
  - Design critique
  - Robustness critique or reanalysis
  - Data provenance / forensic checks
  - Meta-analysis / evidence synthesis
- Important reversals come from post-publication scrutiny